

Experiment 'p_6_7_reverse_array_array' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_7_reverse_array_array

Natural language statement: null

Method signature: p_6_7_reverse_array_array(inputs: array<int>) returns (reversed: array<int>)

Ensures

- `reversed.Length == inputs.Length`
- `forall i :: 0 <= i < reversed.Length ==> reversed[i] == inputs[inputs.Length - 1 - i]`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_7_reverse_array_array(inputs: array<int>) returns (reversed: array<int>)`

The method should respect the following contract:

ensures `reversed.Length == inputs.Length`, ensures forall `i :: 0 <= i < reversed.Length`
 $\hookrightarrow$ `==> reversed[i] == inputs[inputs.Length - 1 - i]`

Produce and show only the Dafny body of this method, including the curly braces that
 $\hookrightarrow$ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 $\hookrightarrow$ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnOPLhHKYguJyfn4cldYrh0Z09dBz

Model: gpt-5.5-2026-04-23

Created at: 1780664267

Finish reason: stop

Usage: 306 tokens in, and 358 tokens out

```
{
  reversed := new int[inputs.Length];
  var i := 0;
  while i < inputs.Length
    invariant 0 <= i <= inputs.Length
    invariant reversed.Length == inputs.Length
    invariant forall j :: 0 <= j < i ==> reversed[j] == inputs[inputs.Length - 1 - j]
  {
    reversed[i] := inputs[inputs.Length - 1 - i];
    i := i + 1;
  }
}
```

Final Program

```
method p_6_7_reverse_array_array(inputs: array<int>) returns (reversed: array<int>)
  ensures reversed.Length == inputs.Length
  ensures forall i :: 0 <= i < reversed.Length ==> reversed[i] == inputs[inputs.
     $\hookrightarrow$  Length - 1 - i]
```

```
{
  reversed := new int[inputs.Length];
  var i := 0;
  while i < inputs.Length
    invariant 0 <= i <= inputs.Length
    invariant reversed.Length == inputs.Length
    invariant forall j :: 0 <= j < i ==> reversed[j] == inputs[inputs.Length - 1 - j]
    {
      reversed[i] := inputs[inputs.Length - 1 - i];
      i := i + 1;
    }
}
```

Total Token Usage

Input tokens: 306

Output tokens: 358

Reasoning tokens: 235

Sum of 'total tokens': 664

Experiment Timings

Overall Experiment started at 1780664266370, ended at 1780664311845, lasting 45475ms (45.48 seconds)

Iteration #1 started at 1780664266375, ended at 1780664311845, lasting 45470ms (45.47 seconds)